

**Quotient Laws of Exponents***Question Bank 3.02*

---

---

*Show your work. Circle your final answer.*

**For questions 1 through 3, show the quotient of powers law, then evaluate.**

1.  $\frac{10^5}{10^3}$

2.  $\frac{8^4}{8^1}$

3.  $\frac{(-6)^6}{(-6)^5}$

4. When dividing two powers that have the same nonzero base, keep the base and \_\_\_\_\_ the exponents.

**For questions 5 and 6, apply the power of a quotient law. Do not simplify the fraction.**

5.  $\left(\frac{-5}{10}\right)^7$

6.  $\left(\frac{3}{-4}\right)^5$

**The zero exponent.**

7. Fill in the missing value in the pattern:  $4^4 = 256$  ,  $4^3 = 64$  ,  $4^2 = 16$  ,  $4^1 = 4$  ,  
 $4^0 =$  \_\_\_\_\_

8. Evaluate  $9^0$  and explain in one sentence why your answer is not 0.

**For questions 9 through 11, rewrite the expression using a law of exponents, name the law, then evaluate.**

9.  $\frac{7.6^5}{7.6^3}$

10.  $\left(\frac{10}{11}\right)^0$

11.  $\left(\frac{0.8}{1.6}\right)^4$

**Working backwards.**

12. Find the value of  $n$  that makes each statement true.

a.  $\frac{3^7}{3^n} = 3^2$

b.  $\left(\frac{5}{2}\right)^n = \frac{125}{8}$

c.  $\frac{6^n}{6^4} = 1$

13. Which expression is equivalent to  $\frac{12^9}{12^3}$  ?

- ☐ A.  $12^6$
- ☐ B.  $12^3$
- ☐ C.  $1^6$
- ☐ D.  $12^{12}$

14. Select ALL of the expressions whose value is 1.

☐ A.  $5^0$

☐ B.  $\frac{3^4}{3^4}$

☐ C.  $(-4)^0$

☐ D.  $1^5$

☐ E.  $\frac{3^4}{3^3}$

☐ F.  $3^0 \cdot 3$